

## **Hook Document: Can You Predict Customer Satisfaction From a Restaurant Review?**

Imagine you are a data analyst working for a restaurant company that owns hundreds of restaurant locations across the country. Every day, customers leave thousands of online reviews discussing their dining experiences. While star ratings provide a quick overview, the written text often explains why customers felt positively or negatively about their experience. Your task is to determine whether customer sentiment can be predicted directly from review text using machine learning and natural language processing (NLP).

Companies increasingly rely on automated sentiment analysis to understand customer satisfaction at scale. Instead of manually reading every review, businesses can use NLP models to quickly identify patterns in customer experiences, recurring complaints, and positive feedback trends. This type of analysis is widely used across industries including restaurants, retail, entertainment, and hospitality.

In this case study, you will work with a real-world restaurant review dataset containing customer-written reviews and star ratings. You will clean and preprocess review text, convert ratings into sentiment categories, vectorize text data, and train classification models capable of predicting sentiment from language patterns.

By the end of this project, you will produce a sentiment analysis report evaluating whether restaurant review language can reliably predict customer sentiment. You will also compare machine learning models and analyze the challenges associated with ambiguous language, class imbalance, and neutral sentiment classification.

Your final deliverable will include:

- A cleaned dataset
- Exploratory analysis of sentiment categories
- A machine learning pipeline for text classification
- Model evaluation metrics and visualizations
- A short written discussion of results and limitations

This project introduces core concepts in natural language processing, supervised machine learning, and text analytics while simulating a real-world business problem.